

- This sample (and your real exam) has 5 questions for 60 marks.
- You are expected to spend approximately two minutes per mark (i.e. your final exam is 2 hours).
- The format of your exam questions is similar to this sample.
- Only UWA approved calculators with stickers are permitted.

Question 1.

(a) Suppose we collect data for a group of students in a machine learning class with two features: x_1 : hours studied and x_2 : undergrad GPA (Grade Point Average).

Our target response is $y = 1$ if the student received an HD (High Distinction), and $y = 0$, otherwise.

We fit a Logistic regression model, where the estimated probability $\hat{p}$ is given by,

$$\hat{p} = h_{\boldsymbol{\theta}}(\mathbf{x}) = \sigma(\boldsymbol{\theta}^\top \mathbf{x}),$$

and the logistic function $\sigma(t)$ is defined as,

$$\sigma(t) = \frac{1}{1 + \exp(-t)}.$$

After fitting our model, we produce estimated coefficients: $\hat{\theta}_0 = -6$, $\hat{\theta}_1 = 0.05$ and $\hat{\theta}_2 = 1$.

How many hours would a student who has an undergrad GPA of 3.5 need to study to have a 50% chance of getting an HD in the class?

(5 marks)

(b) Suppose you are using ridge regression, and you notice that the training and validation errors are almost equal and fairly high.

Would you say that the model suffers from high bias or high variance? Should you increase the regularisation hyperparameter α or reduce it?

(3 marks)

(c) Consider a Linear Regression problem with two features. Based on your visualisation of these 2D features, x_1 and x_2 , on the training set, you noticed that using cubic polynomial features should allow a better model performance.

Write down all the polynomial features in terms of x_1 and x_2 .

(2 marks)

Question 2.

Email spam filtering models often use a bag-of-words representation for emails. In a bag-of-words representation, the descriptive features that describe a document (in our case, an email) represent how often a particular word occurs. One descriptive feature is included for each word in a predefined dictionary. The dictionary is typically defined as the complete set of words in the training dataset.

Table 1 lists the bag-of-words representation for the following five emails and a target feature, SPAM, whether they are spam emails or genuine emails:

- “money, money, money”
- “free money for free gambling fun”
- “gambling for fun”
- “machine learning for fun, fun, fun”
- “free machine learning”

i	Money	Free	For	Gambling	Fun	Machine	Learning	Spam
1	3	0	0	0	0	0	0	True
2	1	2	1	1	1	0	0	True
3	0	0	1	1	1	0	0	True
4	0	0	0	0	3	1	1	False
5	0	1	0	0	0	1	1	False

Table 1: Training instances for the target feature *SPAM*.

Consider the query email: “machine learning for free”.

(a) What is the bag-of-words representation for this query? (2 marks)

(b) What target value would a nearest ($k = 1$) neighbour model using Euclidean distance return for this query? (3 marks)

Recall that the Euclidean distance between data instances x_i and x_j is defined as:

$$D(x_i, x_j) = \sqrt{\sum_{\ell=1}^n (x_i[\ell] - x_j[\ell])^2},$$

where n is the number of features.

(c) What target value would a k -NN model with $k = 3$ and using Euclidean distance return for the same query? Show your working. (5 marks)

Question 3.

(a) Briefly explain what a model's parameter space is, what a feature/input space is, and what model hyperparameters are.

Give a hyperparameter that you would need to consider when designing each of the techniques: Decision Trees, k -NN, and Lasso Regression.

(5 marks)

(b) Explain the differences between multiclass classification and multilabel classification. Give one example of a real-life application in each case. Each example should include a description of the features and the response variable.

(5 marks)

Question 4.

(a) Describe what *random forest classifiers* and *extra-trees* are.

(3 marks)

(b) Briefly explain what *manifold* and *manifold learning* mean.

(3 marks)

(c) Briefly explain what the *kernel trick* (that is commonly used in SVMs) is.

(3 marks)

(d) Explain the difference between *bagging* and *pasting*.

(3 marks)

(e) What is the difference between Supervised Learning and Unsupervised Learning? Give an example for each.

(3 marks)

Question 5.

(a) Briefly explain the training procedure of the SVM classification model.

Your answer should include an explanation about how the margin is defined, what the support vectors are, what training instances are relevant for defining the decision boundary, and what is used as the decision boundary for prediction.

(5 marks)

(b) The following code was used to split a dataset into training and testing sets:

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42
)

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

Implement a ridge regression model using scikit-learn. Grid-search the regularisation strength α with five values. Use k -fold cross-validation with $k = 3$, and refit the best model with the optimal value of α on the whole training set.

Print the best value for α , and calculate the mean squared error on the test set.

(5 marks)

(c) Explain what the OvO “one versus one” strategy is and what it is used for.

(5 marks)

_____ **END OF EXAMINATION PAPER** _____